FORMAL MATHEMATICAL DERIVATION

From Historical Equity Data to Two-Year Monte Carlo Simulations

Three assets · Four stochastic models · 100,000 simulated paths per model and asset

Objective. This report presents the model derivation in a textbook-style mathematical format: each equation is displayed clearly, its symbols are defined, and the sequence from historical returns to simulated prices is made explicit.

Data qualification. The supplied Reliance and Titan training files cover approximately eight years. The supplied Jio Financial training file contains 624 returns from 22 August 2023 to 27 February 2026, so it does not provide eight years of Jio Financial history.

Simulation design. The requested horizon is 494 trading days, approximately two trading years. The intended count is 100,000 paths for each of three assets and four model specifications. This document describes the mathematics; it does not claim that all 12 runs have been independently rerun under one identical seed/configuration.

Notation used throughout

Symbol	Meaning
P■	Observed or simulated closing price at trading day t
r■	Daily continuously compounded (log) return
n	Number of training return observations
T	Forecast horizon; here T = 494 trading days
N	Number of simulated paths per model; here N = 100,000
μ■, s■	Training-sample mean and standard deviation of log returns
u, α■, p■tail	Tail threshold, fitted Pareto exponent, and fitted tail-event probability
ω, a_g, b_g	GARCH variance parameters

1. From closing prices to log returns

Let P■ denote the closing price observed on trading day t. The daily log return is defined by the ratio of consecutive closing prices:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

If the source file already contains a Log_Return column, that series is used after checking that values are numeric and finite. Let r■,…,r■ denote the training returns. Model parameters are estimated from these training observations only; the test period is reserved for out-of-sample evaluation.

For a simulated path k, the cumulative log return is converted back into a price using the final training close P■:

$$P_t^{(k)} = P_0 \exp\left(\sum_{j=1}^{t} r_j^{(k)}\right), \quad t = 1, \ldots, T$$

Here k = 1,…,N identifies the simulation path. The requested forecast horizon is T = 494 trading days and the target number of paths is N = 100,000 for each asset-model pair.

2. Estimating the training-sample location and scale

For the training sample r■,…,r■, the sample mean and unbiased sample variance are:

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} r_t$$

$$\hat{s}^2 = \frac{1}{n-1}\sum_{t=1}^{n} (r_t - \hat{\mu})^2$$

The mean estimates the average daily log return; the standard deviation s■ estimates the daily dispersion of returns. The Gaussian IID model uses these two estimates directly.

3. Model A — Gaussian IID returns

$$r_t^{(k)} = \hat{\mu} + \hat{s} Z_t^{(k)}$$

$$Z_t^{(k)} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

The shocks Z are independent standard normal random variables. This model assumes a constant return distribution through time and does not explicitly represent volatility clustering.

4. Model B — Power-law hybrid IID returns

Let X■ = |r■| be the absolute training return and choose a provisional threshold u = 0.025. Define the set of tail observations I = {t : X■ ≥ u} and let m be its size.

$$\Pr(X > x \mid X \geq u) = \left(\frac{u}{x}\right)^{\alpha}, \quad x \geq u$$

$$\hat{\alpha} = \frac{m}{\sum_{t \in I} \ln(X_t/u)}$$

$$\hat{p}_{\text{tail}} = \frac{m}{n}$$

The simulated return distribution combines empirical resampling of the central body with a Pareto model for tail magnitudes. Tail signs are sampled using the observed sign frequency among historical tail observations.

$$\hat{F}_{\text{hybrid}} = (1 - \hat{p}_{\text{tail}})\hat{F}_{\text{body}} + \hat{p}_{\text{tail}}\hat{F}_{\text{tail}}$$

The Pareto tail is unbounded. A small number of extreme simulated returns can therefore dominate terminal prices; the threshold and fitted exponent must be reported and sensitivity-tested.

5. Model C — GARCH(1,1) with Gaussian innovations

The GARCH model allows the conditional variance to change over time. Returns are decomposed into a conditional mean and an innovation:

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t Z_t$$

$$\sigma_t^2 = \omega + a_g \varepsilon_{t-1}^2 + b_g \sigma_{t-1}^2$$

$$Z_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$$

The parameters are fitted by numerical Gaussian quasi-likelihood. Writing h■ = σ■², the negative log-likelihood objective, omitting constants, is:

$$\mathcal{L}(\mu, \omega, a_g, b_g) = \frac{1}{2} \sum_{t=1}^{n} \left[\ln(h_t) + \frac{(r_t - \mu)^2}{h_t} \right]$$

The variance recursion is constrained to remain positive. The usual covariance-stationarity condition is a_g + b_g < 1; values close to one imply highly persistent conditional volatility and warrant diagnostic review.

6. Model D — GARCH with power-law hybrid innovations

The fourth model retains the GARCH variance recursion but replaces Gaussian innovations with standardized shocks drawn from a fitted hybrid distribution:

$$r_t = \mu + \sigma_t Z_t, \qquad Z_t \sim \widehat{F}_{\text{hybrid, std}}$$

$$\sigma_t^2 = \omega + a_g (r_{t-1} - \mu)^2 + b_g \sigma_{t-1}^2$$

The supplied project script marks the standardized-residual threshold u_z = 2.0 as provisional. This fit is conceptually different from fitting a Pareto tail directly to raw returns; the residual definition, threshold, tail count and exponent should be saved for each asset.

7. Parameter estimates from supplied training files

The table below reports sample estimates and the raw-return Pareto fit at u = 0.025, plus a Gaussian quasi-likelihood GARCH fit computed from the available training returns. GARCH ω is in decimal-return-squared units.

Asset	n	Training dates	μ■	s■	m	p■tail	α■	ω	a_g	b_g
Reliance	1977	2016-10-19 to 2024-10-18	0.00088	0.01782	216	0.109	2.703	2.07e-05	0.098	0.832
Titan	1975	2016-10-25 to 2024-10-22	0.00112	0.01921	251	0.127	2.492	2.01e-06	0.016	0.978
Jio Financial	624	2023-08-22 to 2026-02-27	0.00004	0.02077	111	0.178	2.511	1.15e-05	0.061	0.913

Reliance and Titan training files span approximately eight years. Jio Financial's supplied training file spans only about 2.5 years, so its fitted parameters are based on a shorter sample. The GARCH estimates are numerical fit results, not proof of predictive calibration.

8. Final two-year simulation procedure

For each asset, estimate parameters once from training data and hold them fixed during simulation. For each model and path k, generate T = 494 daily returns according to the appropriate model equation, then reconstruct prices:

$$P_t^{(k)} = P_0 \exp\left(\sum_{j=1}^{t} r_j^{(k)} \right), \quad t = 1, \ldots, 494$$

$$k = 1, \ldots, 100{,}000$$

This yields 100,000 paths per model per asset; 12 model-asset combinations produce 1,200,000 paths overall.

For the terminal-price distribution, compute the empirical median and 5th and 95th percentiles:

$$\widehat{Q}_{0.50} = \text{Quantile}_{0.50} \{P_T^{(k)}\}_{k=1}^{N}$$

$$\widehat{Q}_{0.05}, \quad \widehat{Q}_{0.95}$$

The estimated probability of finishing below the starting price is:

$$\widehat{p}_{\text{below}} = \frac{1}{N} \sum_{k=1}^{N} \mathbf{1}\left(P_T^{(k)} < P_0\right)$$

At N = 100,000, the Monte Carlo standard error for an estimated event probability p■ is approximately √[p■(1–p■)/N], and is at most 0.00158 (0.158 percentage points). This quantifies simulation sampling error only; it does not account for parameter uncertainty or misspecification.

9. What these equations establish — and what they do not

• Mathematical construction: historical log returns are used to estimate the model parameters, simulated returns are generated under each specified law, and cumulative log returns are mapped to prices.

• Numerical output: 100,000 paths approximate each model-implied two-year distribution; terminal-price quantiles and event probabilities are computed from those paths.

• Not established by path count alone: forecast accuracy, correct tail risk, stability to threshold choices, or superiority of one model over another.

• Required evidence for a strong research claim: exact run configuration and saved parameters; held-out test evaluation; proper distributional scores such as CRPS; VaR coverage and independence diagnostics; and threshold-sensitivity analysis.